8.4.14

EE25BTECH11011 - Navya

Question:

Let $\mathbf{P}(x_1, y_1)$ and $\mathbf{Q}(x_2, y_2)$, $y_1 < 0, y_2 < 0$, be the endpoints of the latus rectum of the ellipse $x^2 + 4y^2 = 4$ Then, the equations of the parabolas with latus rectum PQ are

1) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$
2) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$
3) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$
4) $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

Solution:
Consider the ellipse defined by the quadratic form

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \tag{1}$$

where

$$\mathbf{V} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{u} = \mathbf{0}, \quad f = -4. \tag{2}$$

The eigenvalues and eigenvectors of $\mathbf{V}$ are

$$\lambda_1 = 1, \quad \mathbf{p}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \lambda_2 = 4, \quad \mathbf{p}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{3}$$

The eccentricity e of the ellipse is given by

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}. \tag{4}$$

Define the vector

$$\mathbf{n} = \sqrt{\lambda_2}\mathbf{p}_1 = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}. \tag{5}$$

The formula for c is:

$$c = \frac{\mathbf{u}^\top \mathbf{n} \pm \sqrt{e^2(\mathbf{u}^\top \mathbf{n})^2 - \lambda_2(e^2 - 1)(\|\mathbf{u}\|^2 - \lambda_2 f)}}{\lambda_2 e(e^2 - 1)}. \tag{6}$$

Since $\mathbf{u} = \mathbf{0}$, this simplifies to:

$$c = \pm\frac{\sqrt{-\lambda_2(e^2 - 1)(-\lambda_2 f)}}{\lambda_2 e(e^2 - 1)}. \tag{7}$$

Calculate the terms inside the square root:

$$e^2 - 1 = \frac{3}{4} - 1 = -\frac{1}{4}, \tag{8}$$

$$-\lambda_2(e^2 - 1)(-\lambda_2 f) = -4 \times \left(-\frac{1}{4}\right) \times (-4 \times -4) = 16. \tag{9}$$

Thus,

$$c = \pm \frac{\sqrt{16}}{4 \times \frac{\sqrt{3}}{2} \times \left(-\frac{1}{4}\right)} = \mp \frac{2}{\sqrt{3}} \tag{10}$$

The focus $\mathbf{F}$ of the ellipse can be calculated as

$$\mathbf{F} = \frac{ce^2\mathbf{n}}{\lambda_2} = \frac{c \times \frac{3}{4} \times \begin{bmatrix} 2 \\ 0 \end{bmatrix}}{4} = \frac{3c}{8}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{3c}{4}\begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{11}$$

Choosing $c = \frac{2}{\sqrt{3}}$, we get

$$\mathbf{F} = \frac{\sqrt{3}}{2}\begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{12}$$

The endpoints $\mathbf{x}_1, \mathbf{x}_2$ of the latus rectum PQ, where $y_1, y_2 < 0$, satisfy the ellipse equation

$$x^2 + 4y^2 = 4. \tag{13}$$

Substituting $x = \sqrt{3}$, we find

$$y = -\frac{1}{2}, \tag{14}$$

so the endpoints are

$$\mathbf{P} = \begin{bmatrix} \sqrt{3} \\ -\frac{1}{2} \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} -\sqrt{3} \\ -\frac{1}{2} \end{bmatrix} \tag{15}$$

The midpoint (vertex) $\mathbf{V}$ of the latus rectum is

$$\mathbf{V} = \frac{\mathbf{P} + \mathbf{Q}}{2} = \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix}. \tag{16}$$

The length of the latus rectum is

$$|\mathbf{P} - \mathbf{Q}| = 2\sqrt{3} = 4a, \tag{17}$$

which gives

$$a = \pm\frac{\sqrt{3}}{2}. \tag{18}$$

The parabola with vertex $\mathbf{V}$ and parameter a has equation:

$$x^2 = 4a(y - y_0), \tag{19}$$

where $y_0 = -\frac{1}{2}$. Substituting $a = \pm\frac{\sqrt{3}}{2}$, we get

Case 1: $a = +\dfrac{\sqrt{3}}{2}$

$$x^2 - 2\sqrt{3}y = 3 + \sqrt{3}. \tag{20}$$

Case 2: $a = -\dfrac{\sqrt{3}}{2}$

$$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}. \tag{21}$$

Hence, the equations of the required parabolas are

$$\boxed{x^2 - 2\sqrt{3}y = 3 + \sqrt{3} \quad \text{and} \quad x^2 + 2\sqrt{3}y = 3 - \sqrt{3}.} \tag{22}$$